Structure of the workshop

1. Theoretical foundations & The Rule of Three (10:00–11:00)
   - 1.1 Basics: losses & divergences
   - 1.2 Bayesian inference: updating view vs. optimization view
   - 1.3 The rule of three: special cases, modularity & axiomatic foundations
   - 1.4 Q&A/Discussion
2. Optimality of **VI**, **F-VI** suboptimality & **GVI**'s motivation (11:30–12:30)
   - 2.1 VI interpretations: discrepancy-minimization vs constrained optimization
   - 2.2 VI optimality & sub-optimality of F-VI
   - 2.3 GVI as modular and explicit alternative to F-VI
   - 2.4 GVI use cases
   - 2.5 GVI's lower bound interpretation
   - 2.6 Q&A/Discussion
3. **GVI** Applications (14:00 – 15:00)
   - 3.1 Robust Bayesian On-line Changepoint Detection
   - 3.2 Bayesian Neural Networks
   - 3.3 Deep Gaussian Processes
   - 3.4 Other work
   - 3.5 Q&A/Discussion
4. Chalk talk: Consistency & Concentration rates for **GVI** (15:30 – 16:30)
   - 4.1 The role of Γ-convergence
   - 4.2 GVI procedures as ε-optimizers
   - 4.3 Central results
   - 4.4 Q&A/Discussion

Part 2

Part 2: Variational Inference (VI) Optimality & Generalized Variational Inference (GVI)

1. **VI**: Three views
   - 1.1 VI: The lower-bound optimization view
   - 1.2 VI: The discrepancy-minimization view
   - 1.3 VI: The constrained-optimization view
2. Motivating **GVI**
   - 2.1 VI optimality
   - 2.2 F-VI sub-optimality
   - 2.3 GVI: New modular posteriors via $P(\ell_n, D, \mathcal{Q})$
3. **GVI**: use cases
   - 3.1 Robustness to model misspecification
   - 3.2 Prior robustness & Adjusting marginal variances
4. **GVI**: A lower-bound interpretation
   - 4.1 **VI**: lower-bound optimization interpretation
   - 4.2 **GVI**: generalized lower-bound optimization interpretation
5. Discussion

1 Notation & color code

Notation:

(i) $\boldsymbol{\Theta}$ = parameter space, $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ = parameter value

(ii) q, π are **densities** on $\boldsymbol{\Theta}$, i.e. $q, \pi : \boldsymbol{\Theta} \to \mathbb{R}_+$

π = **prior** (i.e., known before data is seen)

q = **posterior** (i.e., known after data is seen)

(iii) $\mathcal{P}(\boldsymbol{\Theta})$ = set of all probability measures on $\boldsymbol{\Theta}$

(iv) $\mathcal{Q}$ = **parameterized** subset of $\mathcal{P}(\boldsymbol{\Theta})$, i.e. $\mathcal{Q}$ = **variational family**

(v) $x_i \overset{iid}{\sim} g(x_i)$; $p(x_i|\boldsymbol{\theta})$ = likelihood model indexed by $\boldsymbol{\theta}$

Color code:

- $P(\ell_n, D, \Pi)$ = loss, Divergence, (sub)space of $\mathcal{P}(\boldsymbol{\Theta})$
- Standard Variational Inference (**VI**)
- Variational Inference based on F-discrepancy (**F-VI**)
- Generalized Variational Inference (**GVI**)

1 VI: Three views

Purpose of section 1: Give three interpretations of **VI**.

(1) Lower-bound on the log evidence (**model selection** view)

(2) Discrepancy-minimization (**F-VI** view)

(3) Constrained optimization (**GVI** view)

1.1 VI: The lower-bound optimization view

Derivation 1: VI maximizes Evidence Lower bound (ELBO):

$$
\begin{aligned}
\log p(x_{1:n}) &= \log\left(\int_{\boldsymbol{\Theta}} p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})d\boldsymbol{\theta}\right) \\
&= \log\left(\int_{\boldsymbol{\Theta}} p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})\frac{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}d\boldsymbol{\theta}\right) \\
&\overset{CoM}{=} \log\left(\mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\frac{p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\right]\right) \\
&\overset{JI}{\geq} \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(\frac{p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\right)\right] \\
&= \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})\right)\right] - \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})\right)\right] \\
&= \mathrm{ELBO}(q)
\end{aligned}
$$

Recipe:

(1) Apply a change of measure (CoM) with parameterized $q(\boldsymbol{\theta}|\boldsymbol{\kappa}) \in \mathcal{Q}$
(2) Lower-bound log-evidence $\log p(x_{1:n})$ with Jensen's Inequality (JI)
(3) Maximizing ELBO = Minimizing 'information loss' due to CoM

Note: Interpretation from model selection (comparing $\log p(x_{1:n}|m)$)

1.2 VI: The discrepancy-minimization view

Derivation 2: VI minimizes KLD between q and p

$$\mathrm{ELBO}(q) = \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})\right)\right] - \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})\right)\right]$$

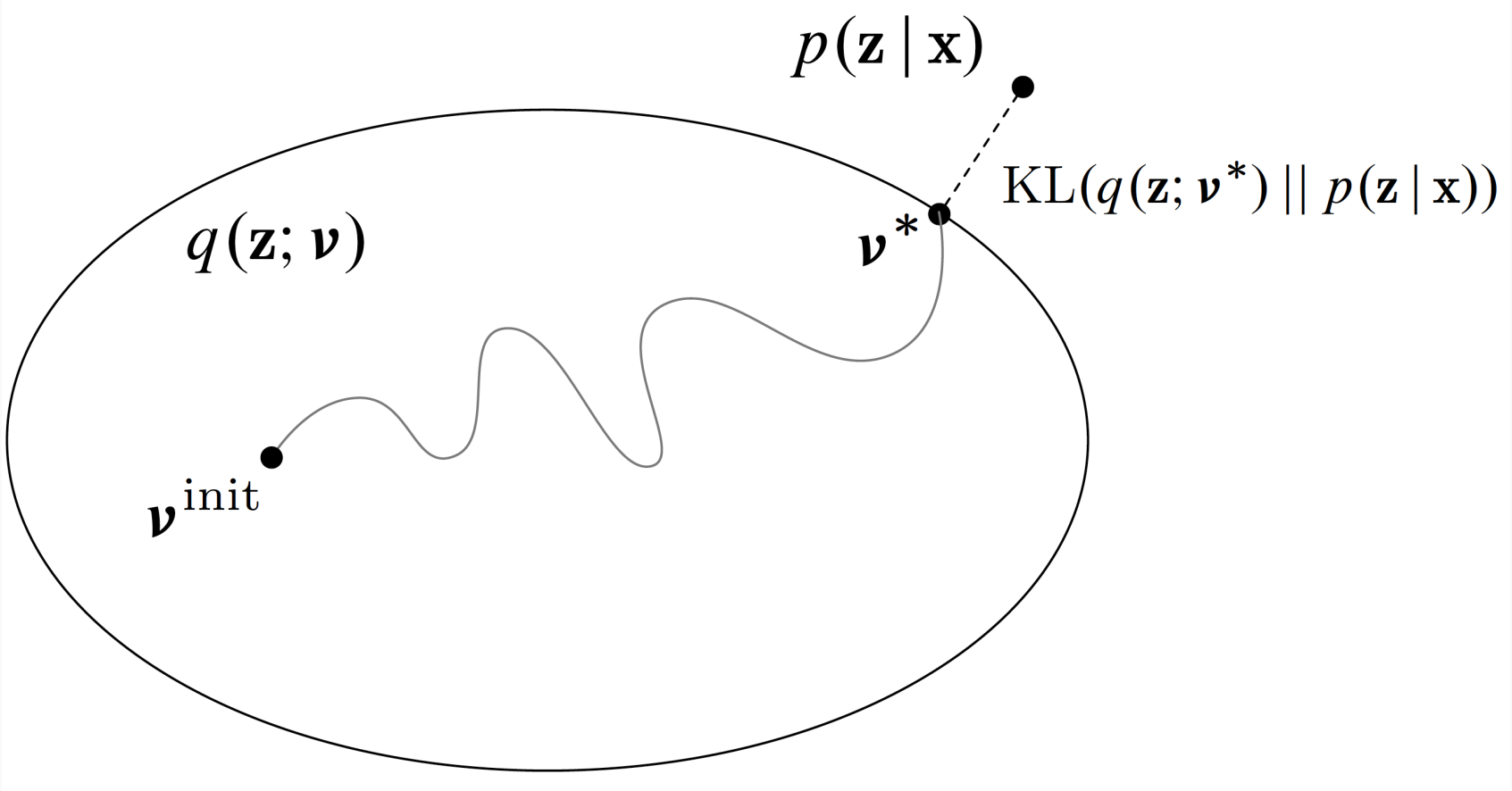
Note 1: $\arg\max_{q\in\mathcal{Q}} \mathrm{ELBO}(q) = \arg\max_{q\in\mathcal{Q}} \left[\mathrm{ELBO}(q) + C\right]$ for any C.
Note 2: Picking $C = -\log p(x_{1:n})$, easy to show that

$$\begin{aligned}\mathrm{ELBO}(q) - \log p(x_{1:n}) &= -\mathbf{KLD}\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})||p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})/p(x_{1:n})\right) \\ &= -\mathbf{KLD}\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})||p(\boldsymbol{\theta}|x_{1:n})\right)\end{aligned}$$

Note 3: So maximizing ELBO = minimizing **KLD** between q & target p!

1.2 VI: The discrepancy-minimization view

Discrepancy-minimization view: VI = approximation minimizing the KLD to $p(\theta|x_{1:n})$. (Inspiration for F-VI methods)



[From *Variational Inference: Foundations and Innovations* (Blei, 2019)]

1.2 VI: discrepancy-minimization view inspires F-VI

F-Variational inference (F-VI): VI based on discrepancy $\boldsymbol{F} \neq \mathrm{KLD}$ (locally) solving

$$q^* = \arg\min_{q \in \mathcal{Q}} \boldsymbol{F}(q \| p) \tag{1}$$

for p = standard Bayesian posterior, e.g.

- $\boldsymbol{F}$ = Rényi's α-divergence (Li and Turner, 2016; Saha et al., 2017)
- $\boldsymbol{F}$ = χ-divergence (Dieng et al., 2017)
- $\boldsymbol{F}$ = operators (Ranganath et al., 2016)
- $\boldsymbol{F}$= scaled AB-divergence (Regli and Silva, 2018)
- $\boldsymbol{F}$ = Wasserstein distance (Ambrogioni et al., 2018)
- $\boldsymbol{F}$ = local reverse KLD (= Expectation Propagation!)

...

1.3 VI: The constrained optimization view

Recall: Minimizing KLD = Maximizing ELBO = Minimizing −ELBO

$$
\begin{aligned}
-\mathrm{ELBO}(q) &= -\mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})\right)\right] + \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})\right)\right] \\
&= -\mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(\prod_{i=1}^{n} p(x_i|\boldsymbol{\theta})\right)\right] + \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(\frac{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}{\pi(\boldsymbol{\theta})}\right)\right] \\
&= \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\Bigg[\underbrace{\sum_{i=1}^{n} -\log p(x_i|\boldsymbol{\theta})}_{=\ell_n(\boldsymbol{\theta}, x_{1:n})}\Bigg] + \mathrm{KLD}\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})||\pi(\boldsymbol{\theta})\right)
\end{aligned}
$$

Observation:

$$
\arg\min_{q\in\mathcal{Q}} -\mathrm{ELBO}(q) \stackrel{!}{=} P(\ell_n, \mathrm{KLD}, \mathcal{Q}) \tag{2}
$$

Conclusion: VI solves problem specified via Rule of Three $P(\ell_n, \mathrm{KLD}, \mathcal{Q})$

1.3 VI: The constrained optimization view

Recall: Minimizing KLD = Maximizing ELBO = Minimizing −ELBO

$$
\begin{aligned}
-\mathrm{ELBO}(q) &= -\mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})\right)\right] + \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})\right)\right] \\
&= -\mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(\prod_{i=1}^{n} p(x_i|\boldsymbol{\theta})\right)\right] + \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(\frac{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}{\pi(\boldsymbol{\theta})}\right)\right] \\
&= \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\Bigg[\underbrace{\sum_{i=1}^{n} -\log p(x_i|\boldsymbol{\theta})}_{=\ell_n(\boldsymbol{\theta}, x_{1:n})}\Bigg] + \mathrm{KLD}\left(q(\boldsymbol{\theta}|\boldsymbol{\kappa})||\pi(\boldsymbol{\theta})\right)
\end{aligned}
$$

Observation:

$$
\arg\min_{q\in\mathcal{Q}} -\mathrm{ELBO}(q) \stackrel{!}{=} P(\ell_n, \mathrm{KLD}, \mathcal{Q}) \tag{3}
$$

Conclusion: VI solves problem specified via Rule of Three $P(\ell_n, \mathrm{KLD}, \mathcal{Q})$

1.3 VI: The constrained optimization view

Key message: VI $=$ $\mathcal{Q}$-constrained version of exact Bayesian inference

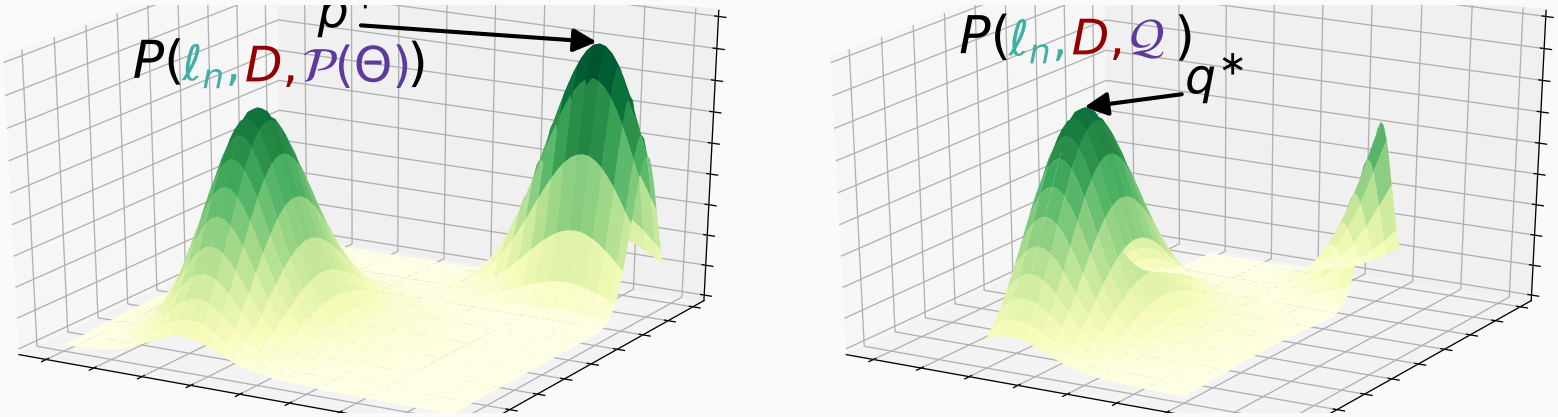


Figure 1 – Left: Unconstrained Bayesian inference. **Right:** $\mathcal{Q}$-constrained Bayesian inference (=VI)

Summary: Three views on VI

(i) Maximize (approximate) $\log p(x_{1:n})$ using $q \in \mathcal{Q}$
$\implies$ **Lower bound view**

(ii) Minimize **F = KLD** discrepancy of q to p
$\implies$ **F-VI** view

(iii) Solve the $\mathcal{Q}$-constrained version of original Bayes problem
$\implies$ **GVI** view

2 Motivating GVI

Purpose of section 2: Relating **VI**, **F-VI** & **GVI**

(1) **VI** optimality

(2) **F-VI** sub-optimality

(3) **GVI**: the best of both worlds

2.1 VI optimality

So VI is the $\mathcal{Q}$-constrained version of original Bayes?!
$\Longrightarrow$ I.e., it is the best you can do in $\mathcal{Q}$!

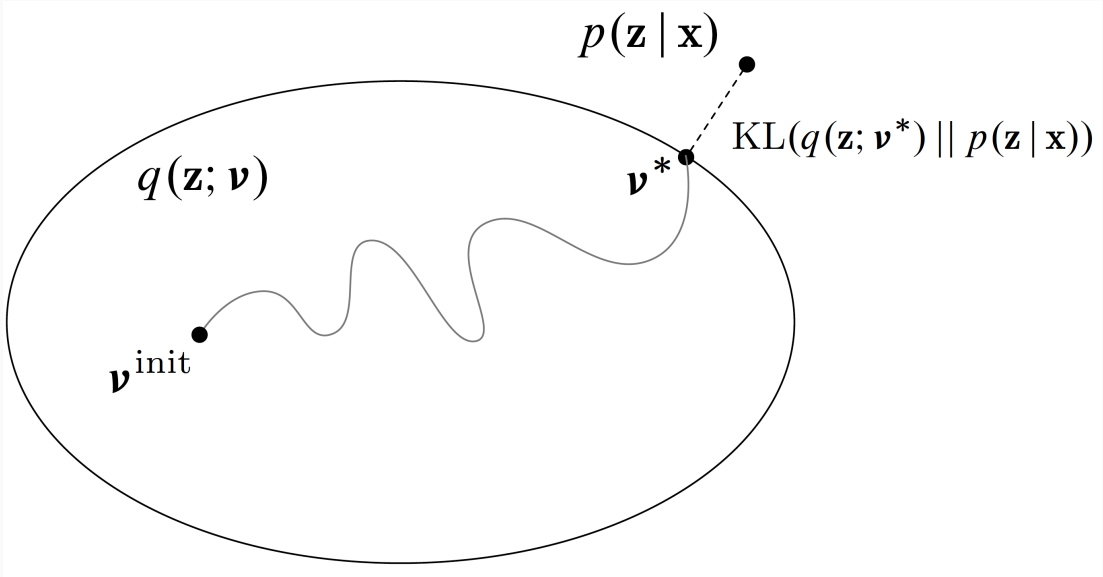
Theorem 1 (VI optimality)
For exact and coherent Bayesian posteriors solving $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$ and a fixed variational family $\mathcal{Q}$, standard VI produces the uniquely optimal $\mathcal{Q}$-constrained approximation to $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$ Having decided on approximating the Bayesian posterior with some $q \in \mathcal{Q}$, VI provides the uniquely optimal solution.

Proof: Many ways to show it, e.g. by contradiction: Suppose there exists a $q' \in \mathcal{Q}$ s.t. q' yields better posterior for original Bayes problem than than $q^{\mathrm{VI}} \Longrightarrow$ contradicts the fact that q^{VI} solves $P(\ell_n, \mathrm{KLD}, \mathcal{Q})$.

2.2 F-VI sub-optimality

Q: If VI is optimal, what about F-VI procedures with $F \neq$ KLD?
$\implies$ For fixed $\mathcal{Q}$, F-VI is sub-optimal relative to $P(\ell_n, \text{KLD}, \mathcal{P}(\Theta))$!

Proof: Same proof applies: Suppose there exists a $q^{\text{F-VI}} \in \mathcal{Q}$ s.t. $q^{\text{F-VI}}$ yields better posterior for original Bayes problem than $q^{\text{VI}} \implies$ contradicts the fact that q^{VI} solves $P(\ell_n, \text{KLD}, \mathcal{Q})$.



Surprising Conclusion: For approximating original Bayesian inference objective as best as one can, F-VI is always worse than standard VI.

2.2 F-VI sub-optimality

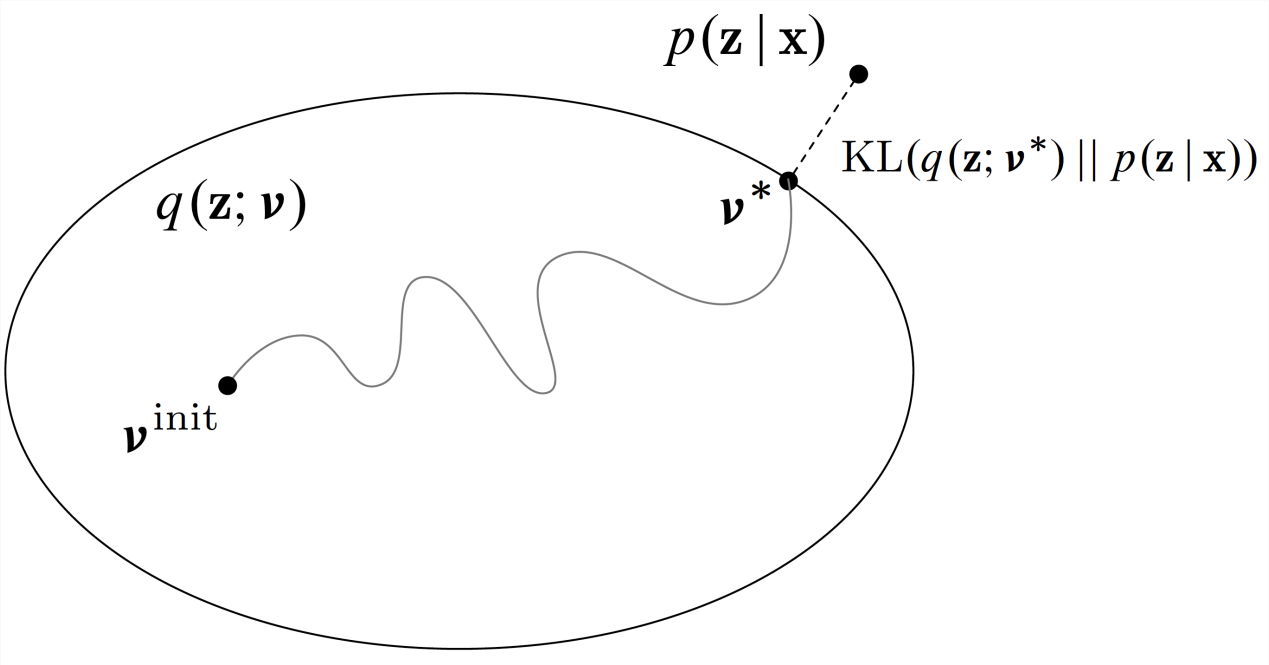
Consequences of F-VI-suboptimality:

(1) If $F \neq \mathrm{KLD}$, **F-VI** violates Axioms 1–4 (in Part 1).

(2) **F-VI** conflates ℓ_n and D (i.e., modularity of $P(\ell_n, D, \boldsymbol{\Pi})$ lost).

(3) Thm: **F-VI** gives worse $\mathcal{Q}$-constrained posterior than standard **VI** (*relative to the standard Bayesian problem* $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$)

Objection! F-VI can produce better posteriors than **VI** in practice!



2.3 GVI: New modular posteriors via $P(\ell_n, D, \mathcal{Q})$

Seeming contradiction:

(1) **VI** is the best approximation to the **standard** Bayesian posterior
(2) **F-VI** often outperforms **VI** (e.g., on test scores)

Resolution: F-VI implicitly targets better Bayesian inference problem than $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$ – but this problem is **non-standard**!
Q: Why not design these non-standard targets explicitly?

(i) Inference should be **optimal** relative to something (i.e. ℓ_n, D & $\mathcal{Q}$)
(ii) Ingredients of non-standard problem should be **interpretable**

$\Longrightarrow$ **Generalized Variational Inference (GVI)**

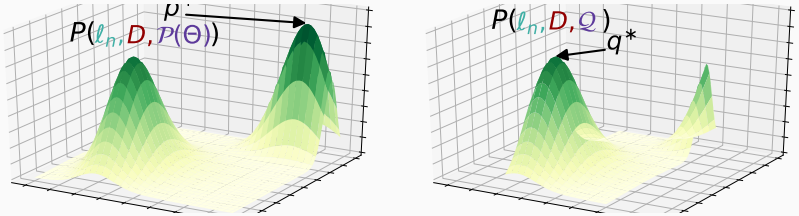


Figure 2 – Left: Unconstrained inference. **Right:** $\mathcal{Q}$-constrained inference (=**VI**)

2.3 GVI: New modular posteriors via $P(\ell_n, D, \mathcal{Q})$

Definition 1 (GVI)

Any Bayesian inference method solving $P(\ell_n, D, \mathcal{Q})$ with admissible choices ℓ_n, D and $\mathcal{Q}$ is a Generalized Variational Inference (**GVI**) method satisfying Axioms 1 – 4.

GVI = combining advantages of **VI** and **F-VI**:

(1) Like **VI**: Has form $P(\ell_n, D, \mathcal{Q})$!
   - (i) satisfies Axioms 1 – 4 (in Part 1);
   - (ii) provably interpretable modularity (**loss**, **uncertainty quantifier**, **admissible posteriors**) (see Part 1)

(2) Like **F-VI**: Targets non-standard posteriors! BUT:
   - (i) without conflating ℓ_n and D
   - (ii) with **explicit** rather than **implicit** changes .

2.3 GVI: New modular posteriors via $P(\ell_n, D, \mathcal{Q})$

Illustration: **F-VI** aims for D, but changes ℓ_n. **GVI** doesn't.

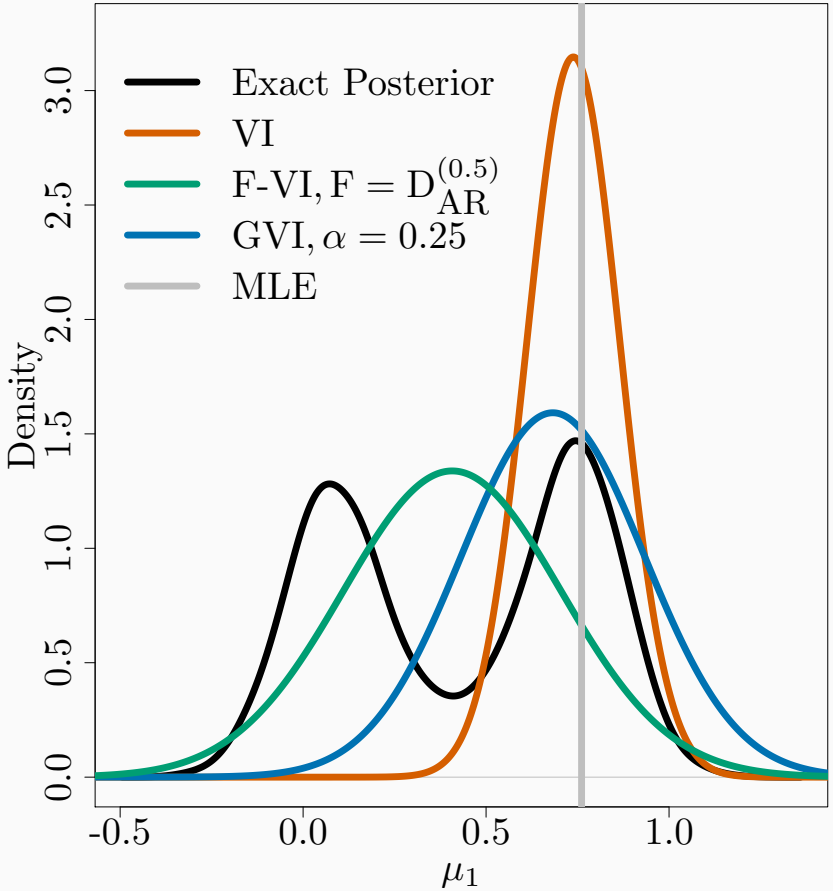


Figure 3 – **Exact**, **VI**, **F-VI** ($F = D_{AR}^{(0.5)}$) and $P(\ell_n, D_{AR}^{(\alpha)}, \mathcal{Q})$ based **GVI** marginals of the location in a 2 component mixture model. Respecting ℓ_n, **VI** and **GVI** provide uncertainty quantification around the most likely value $\widehat{\boldsymbol{\theta}}_n$ via D. In contrast, **F-VI** implicitly changes the loss and has a mode at the locally most *unlikely* value of $\boldsymbol{\theta}$.

Summary: Motivating GVI

(1) **VI** $= P(\ell_n, \mathrm{KLD}, \mathcal{Q})$ is optimal for the original Bayesian problem $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$

(2) **F-VI** $=$ sub-optimal approximation of $P(\ell_n, \mathrm{KLD}, \mathcal{P}(\boldsymbol{\Theta}))$, but can implicitly target non-standard posteriors

(3) **GVI** $= P(\ell_n, D, \mathcal{Q})$: combines the best of both worlds by leveraging the novel constrained-optimization view

3 GVI: What does it do?

Purpose of part 4: Exploring three use cases of **GVI**

(1) Robust alternatives to $\ell(\boldsymbol{\theta}, x_i) = -\log(p(x_i|\boldsymbol{\theta}))$

(2) Prior-robust uncertainty quantification via D

(3) Adjusting marginal variances via D

3.1 GVI: The losses I/III

GVI modularity: The loss ℓ_n

Q1: Why use $\ell_n(\boldsymbol{\theta}, \boldsymbol{x}) = \sum_{i=1}^n -\log(p(x_i|\boldsymbol{\theta}))$?
A: Assuming that the true data-generating mechanism is $\boldsymbol{x} \sim g$,

$$\arg\min_{\boldsymbol{\theta}} \sum_{i=1}^n -\log(p(x_i|\boldsymbol{\theta})) \approx \arg\min_{\boldsymbol{\theta}} \mathbb{E}_g \left[-\log(p(\boldsymbol{x}|\boldsymbol{\theta}))\right]$$
$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_g \left[-\log\left(p(\boldsymbol{x}|\boldsymbol{\theta})\right) + \log(g(\boldsymbol{x}))\right] = \arg\min_{\boldsymbol{\theta}} \mathrm{KLD}(g \| p(\cdot|\boldsymbol{\theta}))$$

Interpretation: $\boldsymbol{-\log\left(p(x_i|\theta)\right)}$ = targeting KLD-minimizing $p(\cdot|\boldsymbol{\theta})$

Q2: Are there other $\boldsymbol{\mathcal{L}^D(p(x_i|\theta))}$ for divergence $\boldsymbol{D}$?
A: Yes! (e.g. Jewson et al., 2018; Futami et al., 2017; Ghosh and Basu, 2016; Hooker and Vidyashankar, 2014)

3.1 GVI: The losses II/III

Q3: Why use other $\boldsymbol{\mathcal{L}^D(p(x_i|\theta))}$?
A: Robustness (for $\boldsymbol{D}$ = a robust divergence) [log/KLD non-robust!]

Robustness recipe: $\alpha/\beta/\gamma$-divergences using generalized log functions
E.g.: β indexes β-divergence ($D_B^{(\beta)}$) via

$$\log_\beta(x) = \frac{1}{(\beta-1)\beta}\left[\beta x^{\beta-1} - (\beta-1)x^\beta\right]$$
$$D_B^{(\beta)}(g||p(\cdot|\boldsymbol{\theta})) = \mathbb{E}_g\left[\log_\beta(p(\boldsymbol{x}|\boldsymbol{\theta})) - \log_\beta(g(\boldsymbol{x}))\right]$$

Note 1: $D_B^{(\beta)} \to \mathrm{KLD}$ as $\beta \to 1$!
Note 2: Admits $D_B^{(\beta)}$-targeting loss as

$$\boldsymbol{\mathcal{L}_p^\beta(\theta, x_i)} = -\frac{1}{\beta-1}p(x_i|\boldsymbol{\theta})^{\beta-1} + \frac{I_{p,\beta}(\boldsymbol{\theta})}{\beta}, \quad I_{p,c}(\boldsymbol{\theta}) = \int p(x|\boldsymbol{\theta})^c dx$$

3.1 GVI: The losses III/III

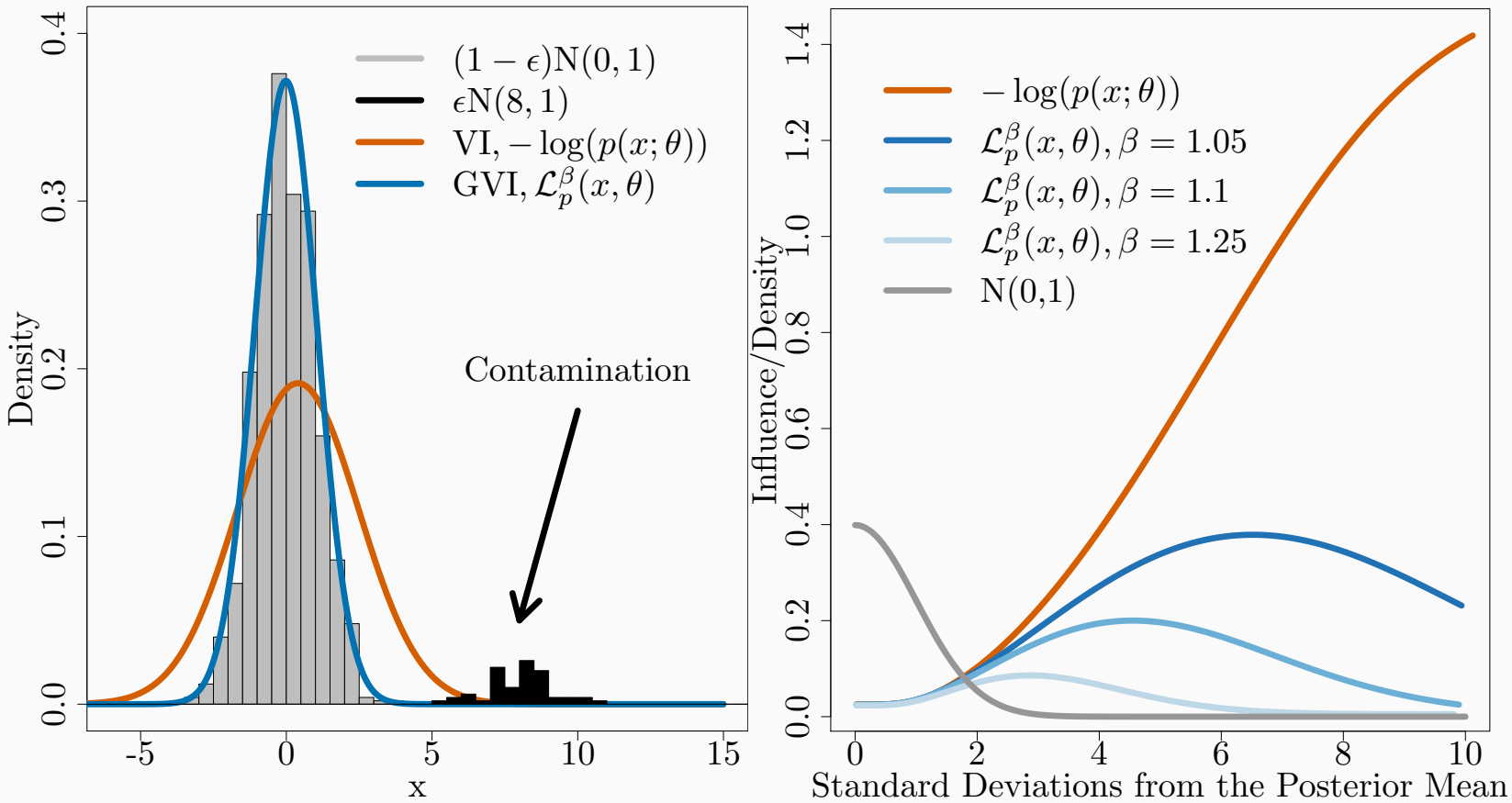


Figure 4 – Left: Robustness against model misspecification. Depicted are posterior predictives under $\varepsilon = 5\%$ outlier contamination using **VI** and $P(\sum_{i=1}^n \mathcal{L}_p^\beta(\boldsymbol{\theta}, x_i), \mathrm{KLD}, \mathcal{Q})$, $\beta = 1.5$. **Right:** From Knoblauch et al. (2018). Influence of x_i on exact posteriors for different losses.

3.1 GVI: Choosing loss hyperparameters I/II

Q: What do losses for β/γ-divergences look like?

$$\mathcal{L}_p^\beta(\boldsymbol{\theta}, x_i) = -\frac{1}{\beta - 1} p(x_i|\boldsymbol{\theta})^{\beta-1} + \frac{l_{p,\beta}(\boldsymbol{\theta})}{\beta}$$

$$\mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i) = -\frac{1}{\gamma - 1} p(x_i|\boldsymbol{\theta})^{\gamma-1} \frac{\gamma}{l_{p,\gamma}(\boldsymbol{\theta})^{\frac{\gamma-1}{\gamma}}}$$

$$l_{p,c}(\boldsymbol{\theta}) = \int p(x|\boldsymbol{\theta})^c dx$$

where $l_{p,c}(\boldsymbol{\theta}) = \int p(x|\boldsymbol{\theta})^c dx$.

Note 1: $\mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i)$ multiplicative & always $< 0 \rightarrow$ **store as log!**
Note 2: Conditional independence $\neq$ additive for $\mathcal{L}_p^\beta(\boldsymbol{\theta}, x_i), \mathcal{L}_p^\gamma(\boldsymbol{\theta}, x_i)$

Q: For losses based on β- or γ-divergence, how to pick β/γ?
Note 3: Good choices for β/γ depend on dimensionality of d
Note 4: In practice, usually best to choose $\beta/\gamma = 1 + \varepsilon$ for some small ε

3.1 GVI: Choosing loss hyperparameters II/II

Q: Any way of choosing hyperparameters?

A: Very much unsolved problem, proposals so far:

- Cross-validation (Futami et al., 2017)
- Via **points of highest influence** (Knoblauch et al., 2018)
- **on-line** updates using **loss-minimization** (Knoblauch et al., 2018)

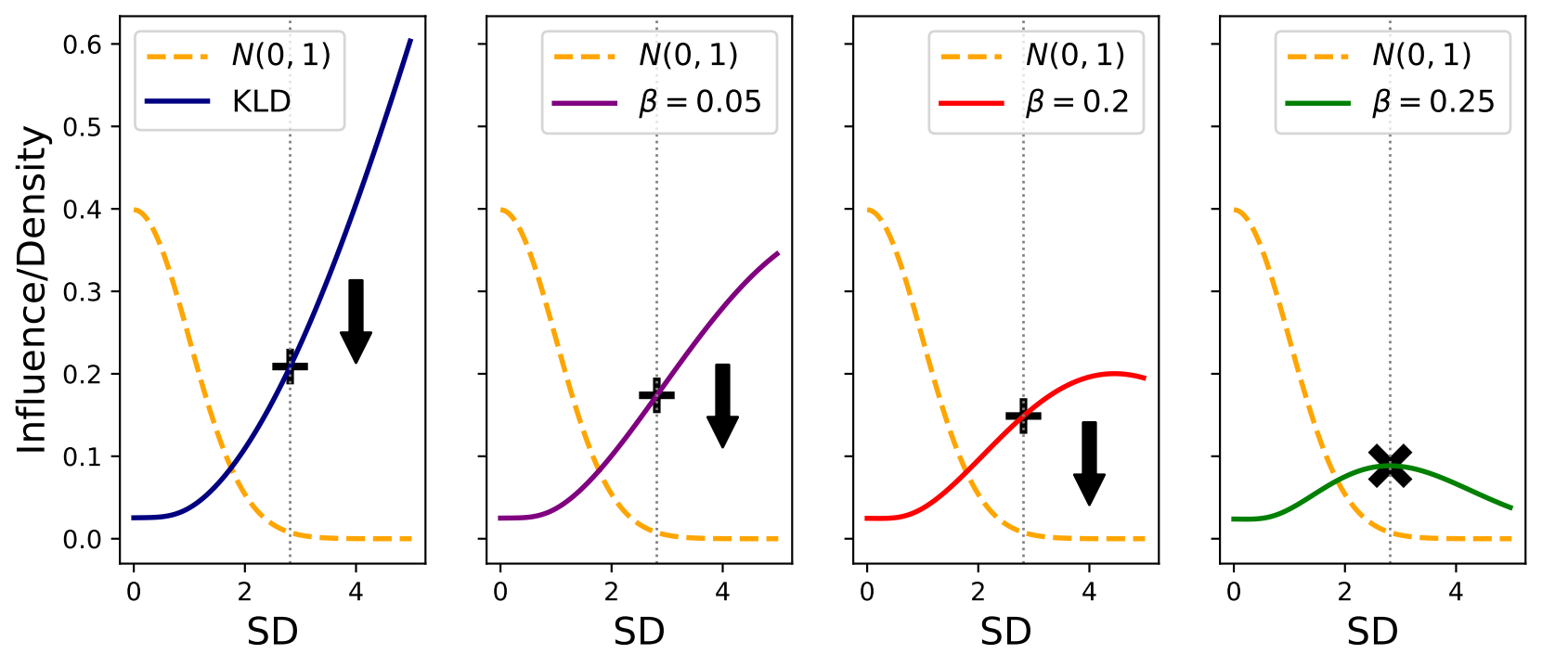


Figure 5 – Illustration of the initialization procedure using **points of highest influence** logic, from left to right.

3.2 GVI: Uncertainty Quantification I/III

GVI modularity: The uncertainty quantifier D

Q: Which VI drawbacks can be addressed via D?

A: Any **uncertainty quantification** properties, e.g.

- Over-concentration (= underestimating marginal variances)
- Sensitivity to badly specified priors
- ...

3.2 GVI: Uncertainty Quantification II/III

Example 1: GVI can fix over-concentrated posteriors

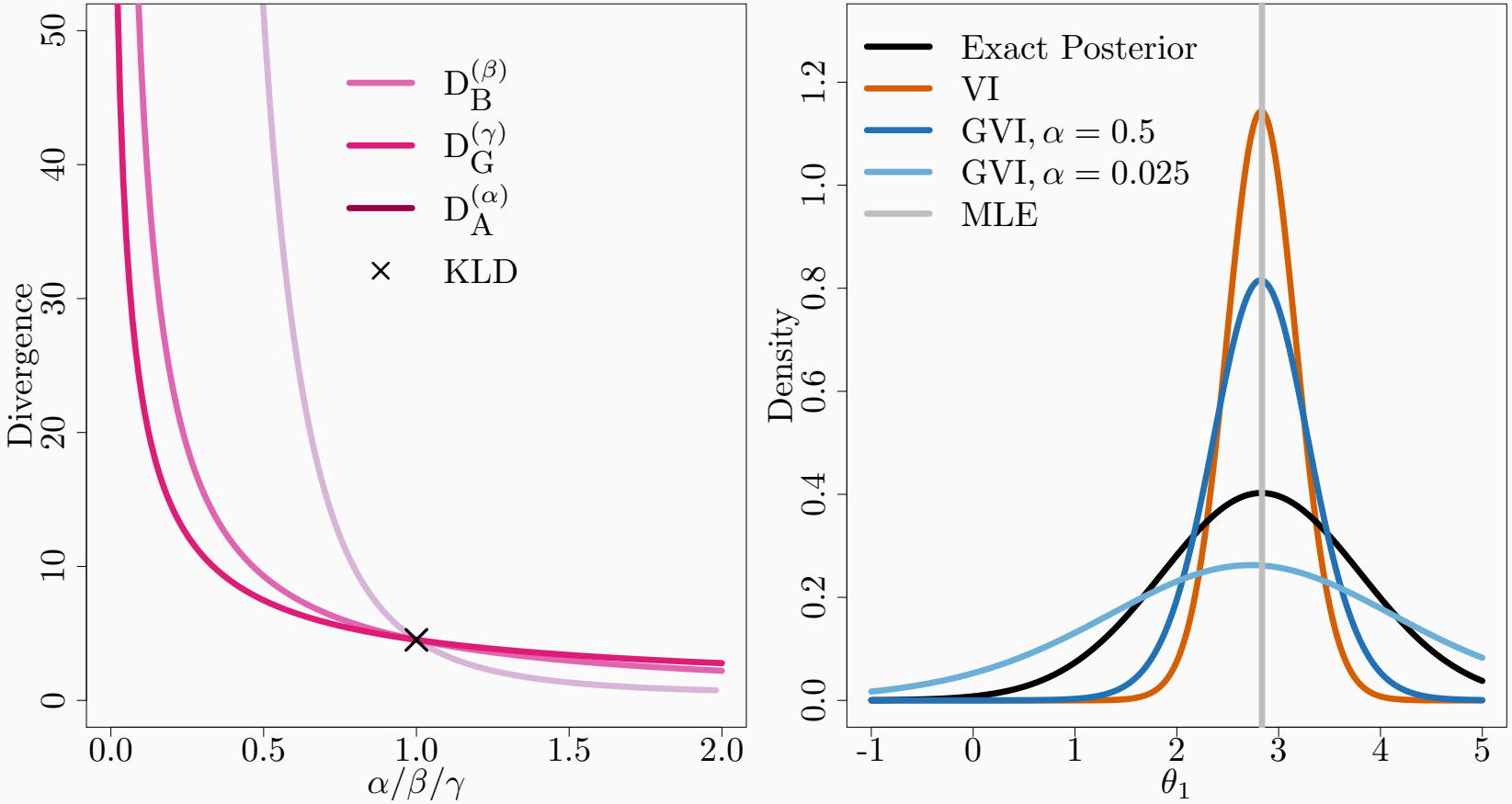


Figure 6 – Left: Magnitude of the penalty incurred by $D(q||\pi)$ for different uncertainty quantifiers D and fixed densities π, q. **Right**: Using $D_{AR}^{(\alpha)}$ with different choices of α to "customize" uncertainty.

3.2 GVI: Uncertainty Quantification III/III

Example 2: Avoiding prior sensitivity

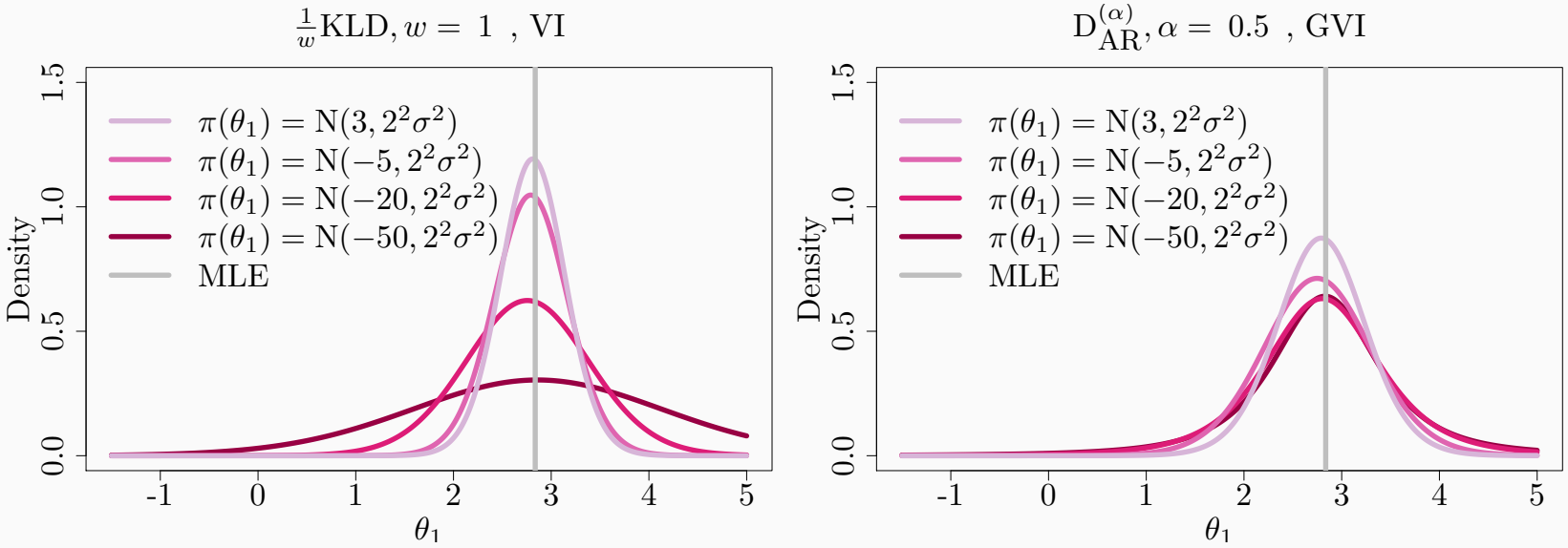


Figure 7 – Prior sensitivity with **VI** (**left**) vs. prior robustness with **GVI** (**right**). Priors are more badly specified for **darker** shades.

Summary: GVI, three use cases

Summary: some **GVI** applications include

(1) Robustness to model misspecification ($=$ adapting ℓ_n)

(2) "Customized" marginal variances ($=$ adapting D)

(3) Prior robustness ($=$ adapting D)

4 GVI: A lower-bound interpretation

Purpose of section 4: Use theory of generalized Bayesian posteriors for lower-bound interpretation of **GVI**

(1) Deriving a generalized lower bound

(2) **VI** = special case holding without slack

(3) **GVI**: Some interpretable bounds with $\mathsf{D} = D_{AR}^{(\alpha)}, D_{G}^{(\gamma)}, D_{B}^{(\beta)}$

4.1 VI lower bound interpretation

Recall: VI also interpretable as optimizing a lower bound on log evidence!

$$\log p(x_{1:n}) \overset{JI}{\geq} \mathbb{E}_{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\left[\log\left(\frac{p(x_{1:n}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{q(\boldsymbol{\theta}|\boldsymbol{\kappa})}\right)\right] = \mathrm{ELBO}(q)$$

Recall: We also saw that VI $= P(\ell_n, D, \mathcal{Q})$ minimizes

$$-\mathrm{ELBO}(q) = -\log p(x_{1:n}) + \mathrm{KLD}(q(\boldsymbol{\theta}|\boldsymbol{\kappa})||p(\boldsymbol{\theta}|x_{1:n})) \tag{4}$$

Thus: max. ELBO = min. KLD = max. lower-bound on $\log p(x_{1:n})$
Q: Can we extend this logic to GVI?
A: Yes, but we need to introduce a few things first . . .

4.1 VI lower bound interpretation

Introduce: Generalized Bayes posterior (Bissiri et al., 2016): $P(\ell_n, D, \mathcal{P}(\Theta))$

$$q^*_{\ell_n}(\boldsymbol{\theta}) \propto \pi(\boldsymbol{\theta}) \exp\{-\ell_n(\boldsymbol{\theta}, x_{1:n})\}$$

Introduce: Generalized evidence (corresponding to $q^*_{\ell_n}$)

$$p_{[\ell_n]}(x_{1:n}) = \int_{\Theta} q^*_{\ell_n}(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

Introduce: Functions L^D, E^D, T^D depending on D:

$$L^D, E^D : \mathbb{R} \to \mathbb{R} \quad \text{(loss and evidence maps)}$$
$$T^D : \mathcal{Q} \to \mathbb{R} \quad \text{(approximate target map)}$$

For VI: $E^{\mathrm{KLD}}(x) = L^{\mathrm{KLD}}(x) = x$, $T^D(q) = \mathrm{KLD}(q || q^*_{\ell_n})$ recovers:

$$-\mathrm{ELBO}(q) = E^{\mathrm{KLD}}\left(-\log p_{[L^{\mathrm{KLD}}(\ell_n)]}(x_{1:n})\right) + T^{\mathrm{KLD}}(q)$$

For GVI: With D based on generalized log, one recovers

$$L(q) \geq E^D\left(-\log p_{[L^D(\ell_n)]}(x_{1:n})\right) + T^D(q)$$

4.2 GVI lower bound interpretation

For VI: $E^{\mathrm{KLD}}(x) = L^{\mathrm{KLD}}(x) = x$, $T^D(q) = \mathrm{KLD}(q || q^*_{\ell_n})$ recovers:

$$-\mathrm{ELBO}(q) = E^{\mathrm{KLD}}\left(-\log p_{[L^{\mathrm{KLD}}(\ell_n)]}(x_{1:n})\right) + T^{\mathrm{KLD}}(q)$$

For GVI: With D based on generalized log, one recovers

$$L(q) \geq E^D\Big(\underbrace{-\log p_{[L^D(\ell_n)]}(x_{1:n})}_{\substack{\text{generalized negative log evidence;} \\ L^D \text{ maps } \ell_n \text{ into a new loss}}}\Big) + \underbrace{T^D(q)}_{\text{Approximate target}}$$

Note: VI = special case which holds **with equality**

$\Longrightarrow$ For VI, approximate target $T^{\mathrm{KLD}}(q)$ = exact target!

4.2 GVI lower bound interpretation

For GVI: With D based on generalized log, one recovers

$$L(q) \geq E^D\Big(\underbrace{-\log p_{[L^D(\ell_n)]}(x_{1:n})}_{\substack{\text{generalized negative log evidence;} \\ L^D \text{ maps } \ell_n \text{ into a new loss}}} \Big) + \underbrace{T^D(q)}_{\text{Approximate target}}$$

Example: Similar results/decompositions for $D_{AR}^{(\alpha)}$, $D_B^{(\beta)}$, $D_G^{(\gamma)}$.

Rényi's α-divergence ($D_{AR}^{(\alpha)}$) for $\alpha > 1$ gives

$$E^{D_{AR}^{(\alpha)}}(z) = \frac{1}{\alpha} \cdot z,$$
$$L^{D_{AR}^{(\alpha)}}(\ell_n(\boldsymbol{\theta}, x_{1:n})) = \alpha \cdot \ell_n(\boldsymbol{\theta}, x_{1:n}),$$
$$T_{D_{AR}^{(\alpha)}}(q) = \frac{1}{\alpha}\mathrm{KLD}(q || q^*_{L^{D_{AR}^{(\alpha)}}(\ell_n)}),$$

so putting it together one finds that for $D = D_{AR}^{(\alpha)}$ with $\alpha > 1$,

$$L(q) \geq -\frac{1}{\alpha} \log p_{\alpha\ell_n}(x_{1:n}) + \frac{1}{\alpha}\mathrm{KLD}(q || q^*_{\alpha\ell_n})$$

Note: Above is $\frac{1}{\alpha}$-scaled version of the ELBO with the loss $\alpha\ell_n$!

$\Longrightarrow$ **approximately** minimizes $\log p_{\alpha\ell_n}(x_{1:n})$ of α-power posterior in $\mathcal{Q}$

4.2 GVI lower bound interpretation

Q: Two different problems, same objective?!

ℓ_n-loss **GVI** with D$=D_{AR}^{(\alpha)}$:

$$L(q) \geq -\frac{1}{\alpha}\log p_{\alpha\ell_n}(x_{1:n}) + \frac{1}{\alpha}\mathrm{KLD}(q||q^*_{\alpha\ell_n})$$

α-Power ℓ_n-loss **VI**:

$$\frac{1}{\alpha}ELBO(q) = -\frac{1}{\alpha}\log p_{\alpha\ell_n}(x_{1:n}) + \frac{1}{\alpha}\mathrm{KLD}(q||q^*_{\alpha\ell_n})$$

No! Slack term!

$$\frac{1}{\alpha}ELBO(q) \stackrel{!}{=} \ldots$$

$$L(q) \stackrel{!}{=} \text{Slack(q)} - \frac{1}{\alpha}\log p_{\alpha\ell_n}(x_{1:n}) + \frac{1}{\alpha}\mathrm{KLD}(q||q^*_{\alpha\ell_n})$$

4.2 GVI lower bound interpretation

Q: What does the slack term represent? $\Longrightarrow$ **Prior robustness!!!**

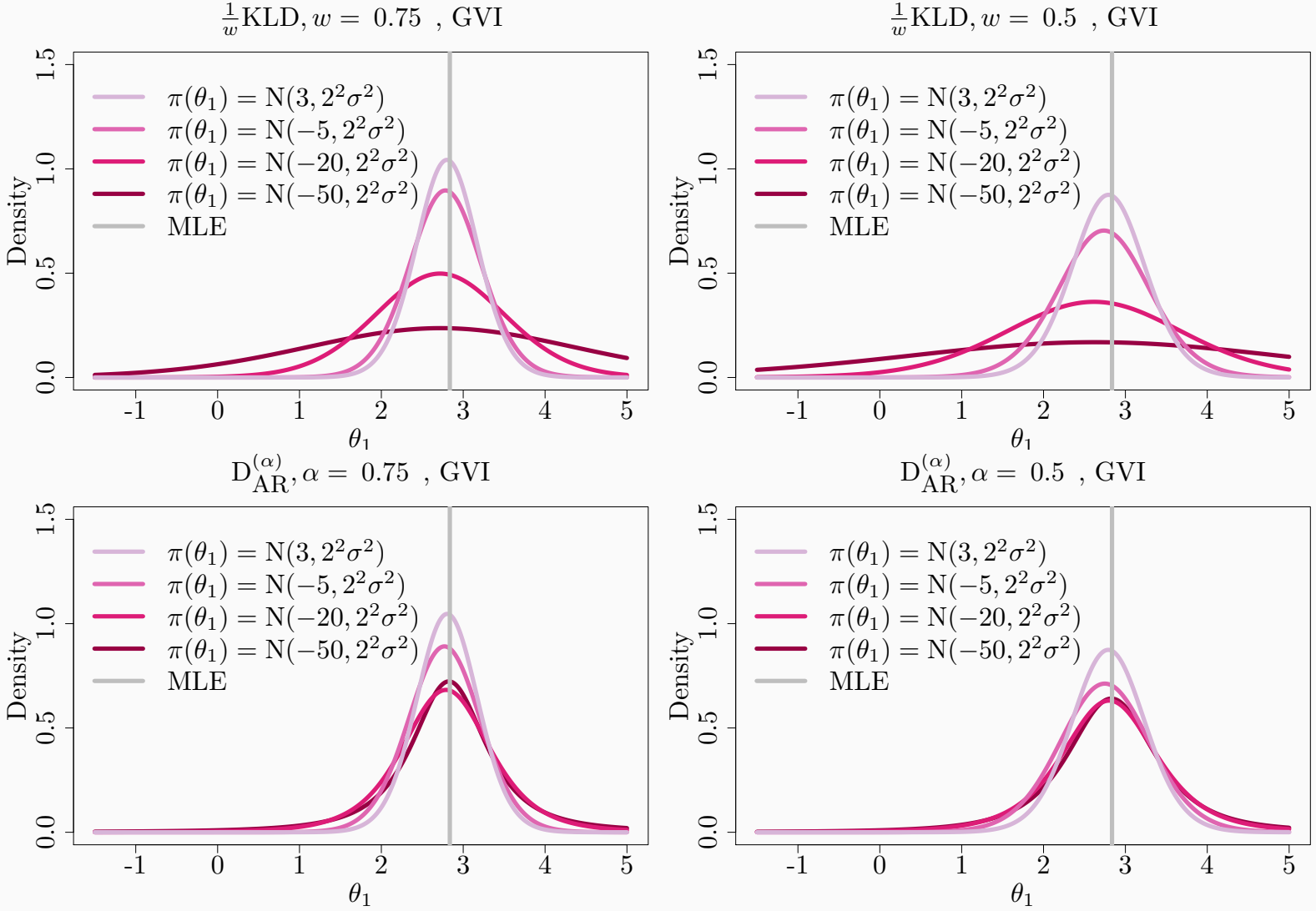


Figure 8 – Power Bayes **VI not** prior-robust – but **GVI** is!

Summary: GVI, a lower-bound interpretation

(1) One can bound **GVI** objectives via a generalized log evidence term $E^D\left(-\log p_{[L^D(\ell_n)]}(x_{1:n})\right)$ plus an approximate target term $T^D(q)$

(2) **Idea:** Minimizing **GVI** objective = approximately minimizing target term $T^D(q)$

(3) **Finding 1:** This is simlar to approximatly minimizing **VI** objective of a power posterior

(4) **Finding 2:** The **slack** term (i.e. what makes the bound approximate) accounts for the **prior robustness**

Discussion / Q&A

If you have questions that you think are stupid, I am sure they are not – here are some questions to compare against that people actually asked on Reddit:

- *When did 9/11 happen?*
- *Can you actually lose weight by rubbing your stomach?*
- *What happens if you paint your teeth white with nail polish?*
- *Is it okay to boil headphones?*
- ...

Main References i

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Appendix: Choosing D for conservative marginals I/II

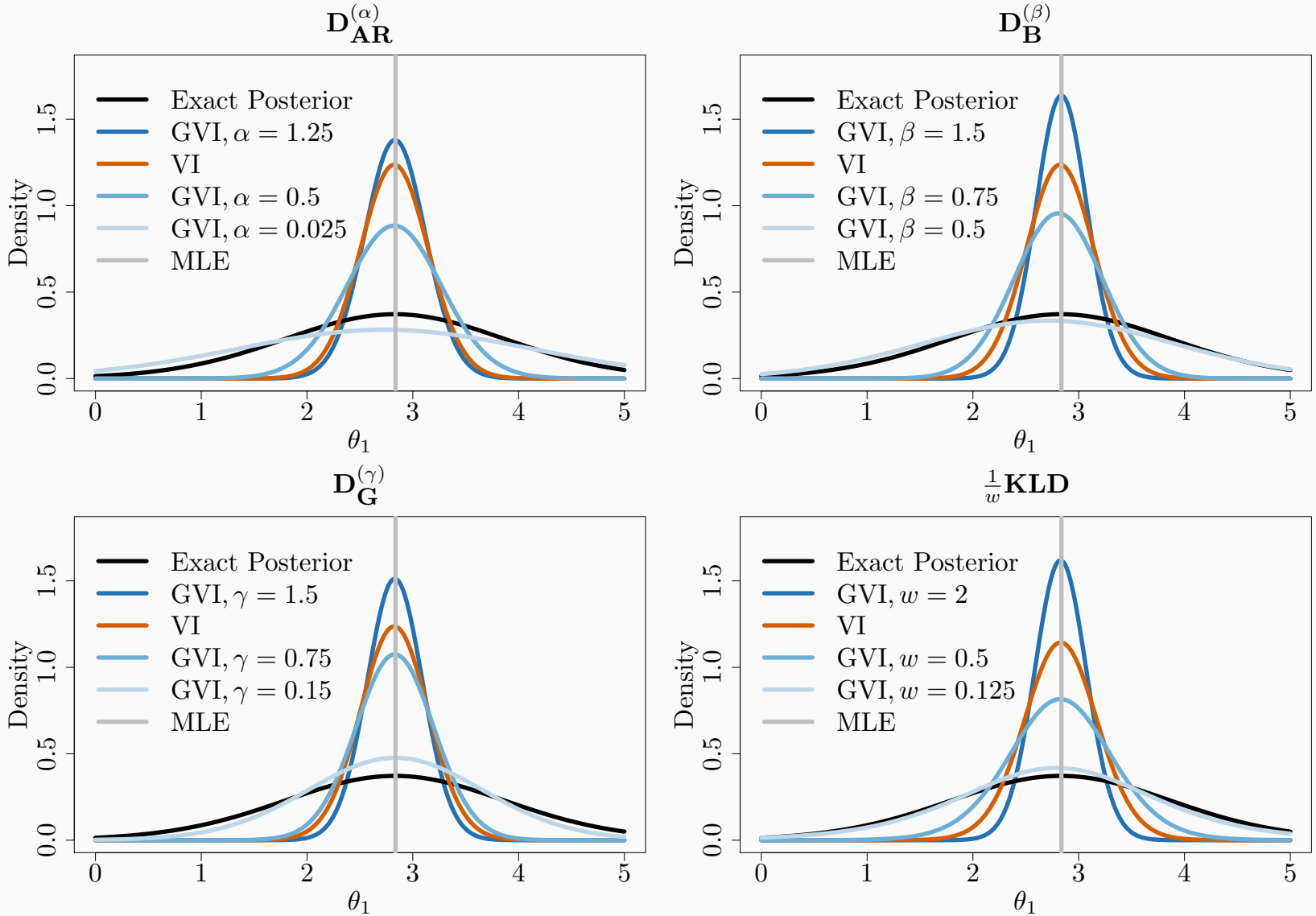

Figure 9 – Marginal VI and GVI posterior for a Bayesian linear model under the $D_{AR}^{(\alpha)}$, $D_{B}^{(\beta)}$, $D_{G}^{(\gamma)}$ and $\frac{1}{w}\mathrm{KLD}$ uncertainty quantifier for different values of the divergence hyperparameters.

Appendix: Choosing D for conservative marginals II/II

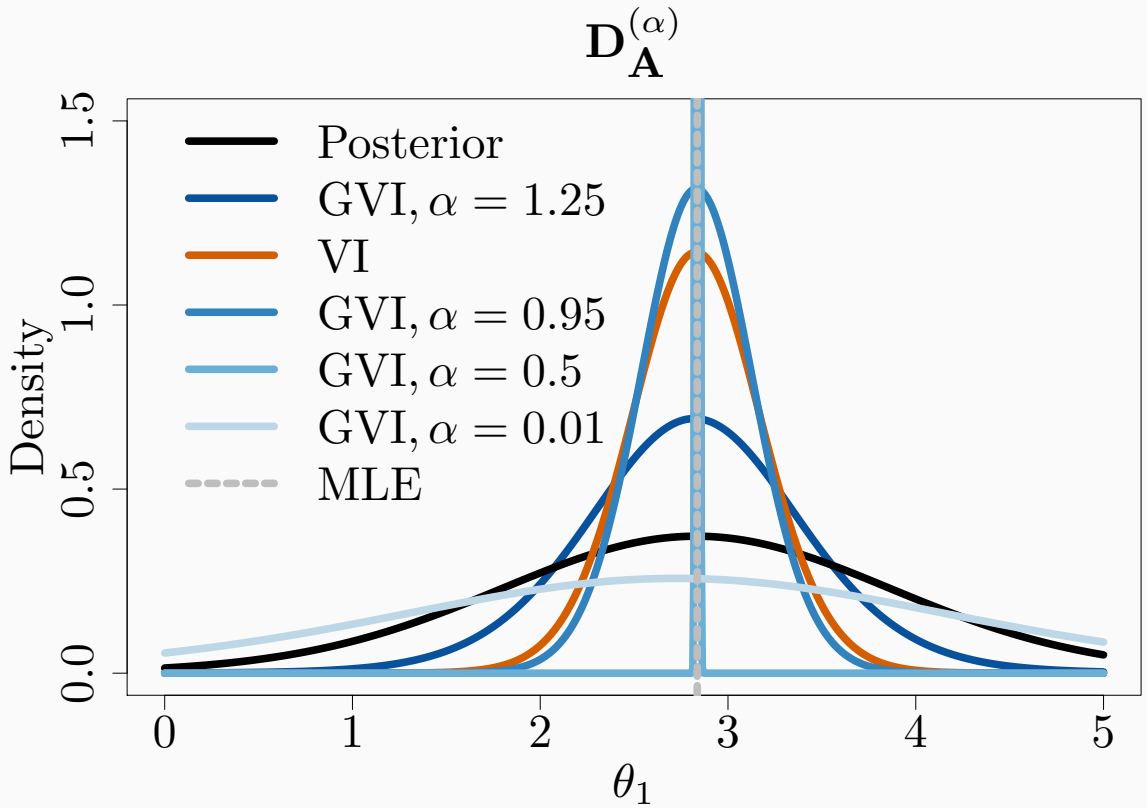

Figure 10 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under the $D_A^{(\alpha)}$ uncertainty quantifier. The boundedness of the $D_A^{(\alpha)}$ causes **GVI** to severely over-concentrate if α is not carefully specified.

Appendix: Choosing D for prior robustness I/IV

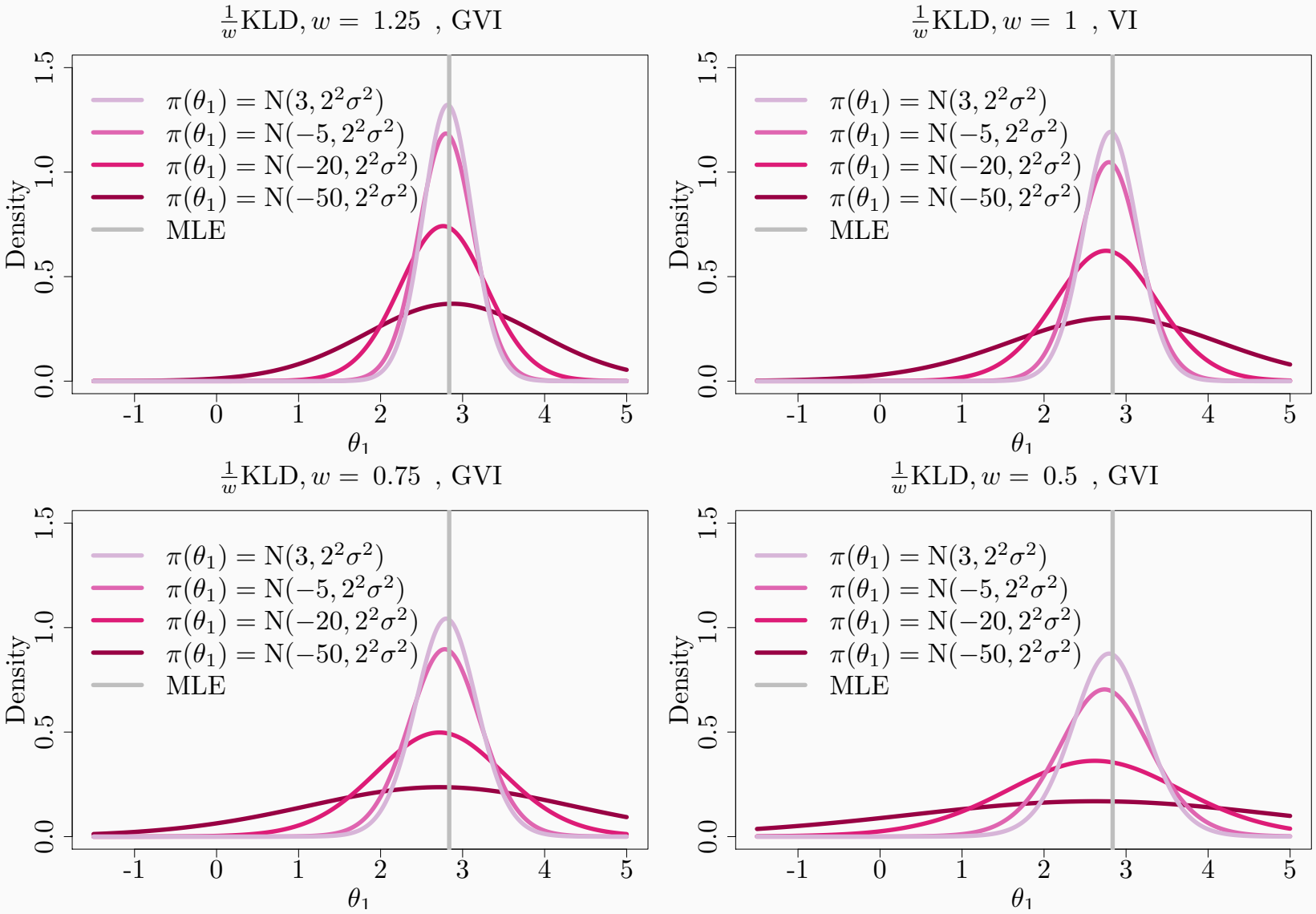

Figure 11 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = \frac{1}{w}\mathrm{KLD}$ as the uncertainty quantifier.

Appendix: Choosing D for prior robustness II/IV

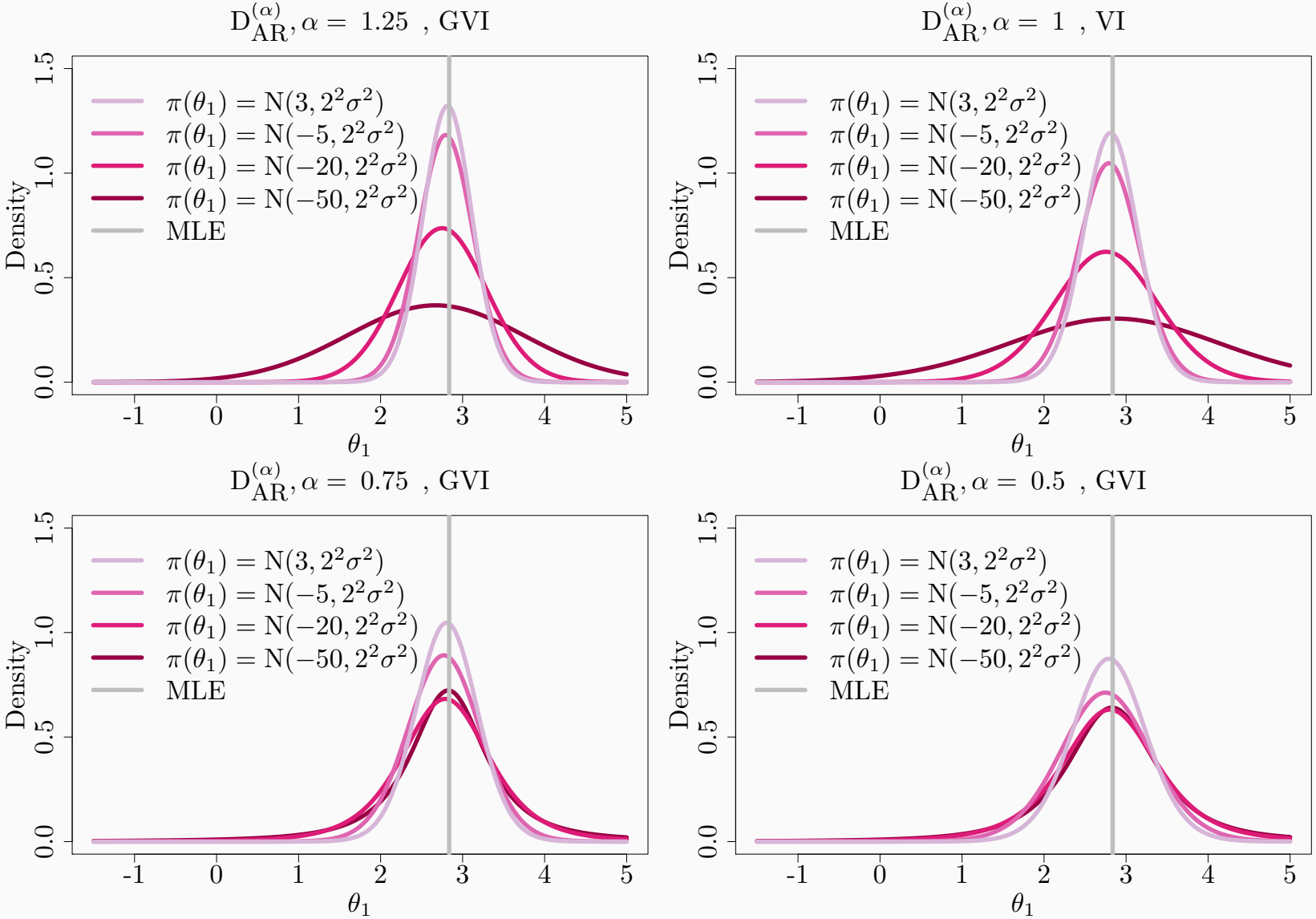

Figure 12 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = D_{AR}^{(\alpha)}$ as the uncertainty quantifier.

Appendix: Choosing D for prior robustness III/IV

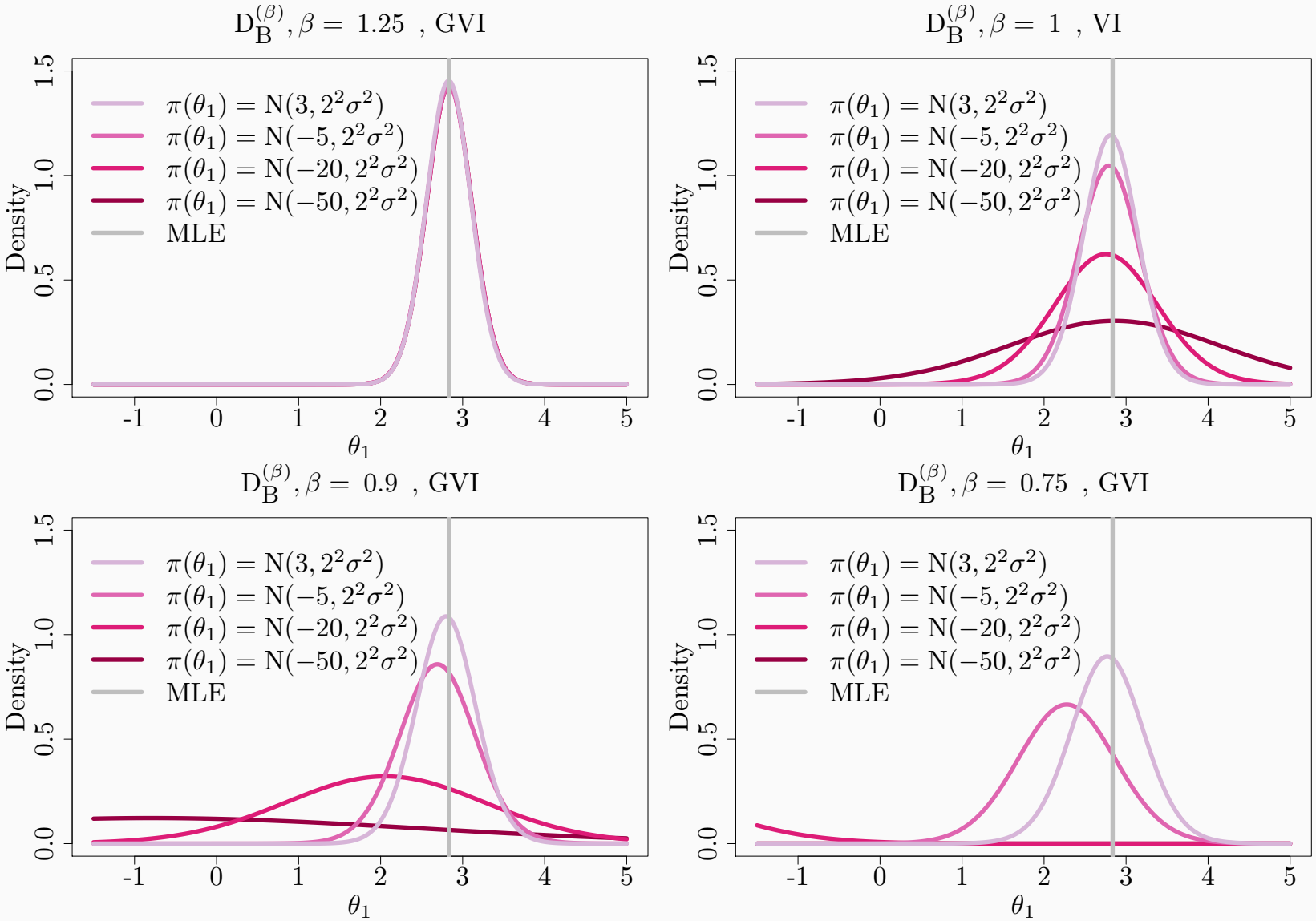

Figure 13 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = D_B^{(\beta)}$ as the uncertainty quantifier.

Appendix: Choosing D for prior robustness IV/IV

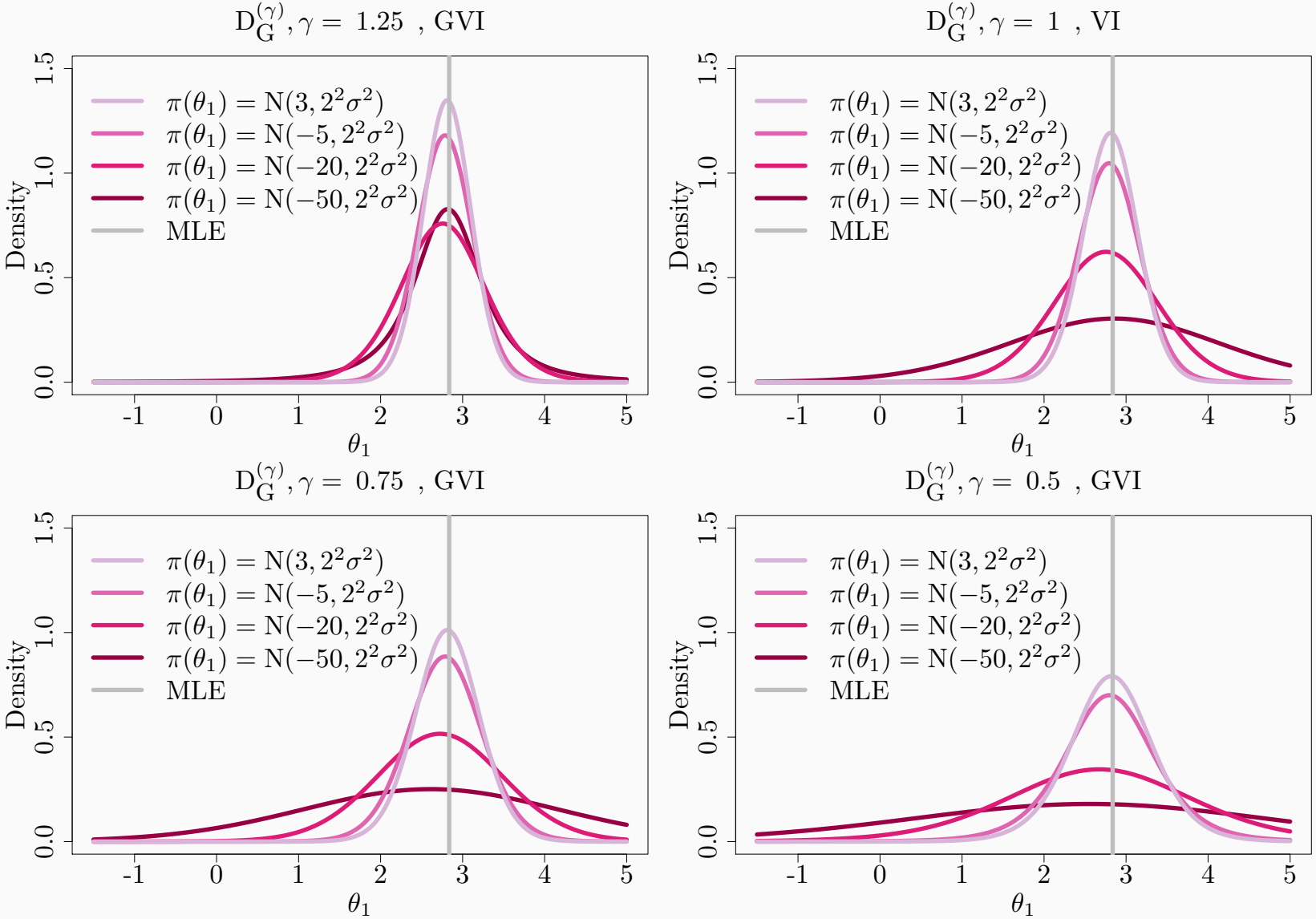

Figure 14 – Marginal **VI** and **GVI** posterior for a Bayesian linear model under different priors, using $D = D_G^{(\gamma)}$ as the uncertainty quantifier.